

ANANYA SRIDHAR

AI/ML Engineer — Applied AI, Speech AI, Computer Vision & Robotics

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GitHub: github.com/darayavahoush · Portfolio: darayavahoush.github.io · vaaksudhi.manaslearning.com · vaakify.manaslearning.com · LinkedIn: linkedin.com/in/ananya-sridhar-ai

Languages: English, Hindi, Tamil

PROFESSIONAL SUMMARY

Full-Stack AI/ML Engineer with a proven track record of architecting and deploying end-to-end production systems across **Speech AI, Agentic Systems (LangGraph, RAG), Computer Vision, and Autonomous Robotics**. Experienced in taking complex machine learning architectures—from physics-backed MuJoCo simulations and vision transformer optimal transport to domain-fine-tuned legal LLMs—into scalable, low-latency, client-facing applications. Skilled in high-performance backends (FastAPI, WebSockets, PostgreSQL), modern web frameworks (React, Vite, Next.js, Three.js), and edge/CPU optimization.

TECHNICAL SKILLS

- Languages:** Python (Proficient), C++, C, SQL, JavaScript (ES6+), TypeScript
- AI / Deep Learning:** PyTorch, Hugging Face (Transformers, PEFT, Accelerate), LangGraph, LangChain, Dense RAG, ChromaDB, Reinforcement Learning (DRL, PPO, Q-Learning), MCTS, A* Pathfinding, Scikit-Learn
- Speech & Audio AI:** Faster-Whisper, OpenAI Whisper, Vosk, Coqui TTS, Kokoro ONNX, Librosa, Praat-Parselmouth, Epitran (G2P), Acoustic Feature Extraction, Phoneme Verification
- Computer Vision:** OpenCV, MediaPipe Tasks Vision, DINOv2 (Vision Transformers), Optimal Transport (POT), Background Subtraction, Multi-Object Centroid Tracking, Edge Detection
- Robotics & Simulation:** MuJoCo 3D Physics Simulation, Gymnasium, Sensor Fusion (Extended Information Filter - EIF), Three.js, React Three Fiber, Forward & Inverse Kinematics, Lidar Motion Planning
- Backend & Cloud:** FastAPI, Uvicorn, PostgreSQL, SQLite, Supabase, Azure Cloud, Docker, WebSockets, RESTful APIs, Alembic
- Frontend & Tooling:** React.js, Vite, Next.js (App Router), Tailwind CSS, HTML5 Canvas, Git, Linux, Bash

PROFESSIONAL EXPERIENCE

AI/ML ENGINEER, FULL-STACK | MANAS LEARNING

Mar 2026 — Present | Remote | Mumbai, India

- End-to-End Platform Architecture:** Architected and deployed full-stack pediatric speech therapy platforms (vaaksudhi.manaslearning.com and vaakify.manaslearning.com) serving clinical assessments across 8 Indian languages.
- LangGraph Speech Assessment Pipeline:** Engineered a real-time pronunciation scoring engine chaining Faster-Whisper, Vosk, and Kokoro/Coqui TTS with phoneme-level acoustic verification (Librosa, Epitran G2P) for instant articulatory feedback across submodules (BreathQuest, VaakMirror, Chime, VoiceHurdleRace).
- Reinforcement Learning Adaptive Engine:** Designed and implemented a DRL/Q-learning adaptive difficulty controller that dynamically calibrates gameplay difficulty per child based on rolling pronunciation accuracy and response latency.
- Visual Biofeedback & Safety:** Integrated MediaPipe facial landmark detection for real-time oral-motor exercise tracking; implemented COPPA-compliant parental consent, secure session tokens, role-based dashboards, and automated clinical progress reports.
- Real-World Impact:** Scaled clinical speech therapy assessments across 8 Indian languages with sub-300ms latency, enabling therapists to deliver automated articulatory error analysis to hundreds of children.

AI/ML ENGINEER | NIVESH AI

Jun 2026 — Present | Mumbai, India

- Enterprise Speech AI Infrastructure:** Built end-to-end audio ingestion and intelligence pipelines deployed on Azure for a wealth management platform serving 10,000+ mutual fund distributors (MFDs).
- Domain-Specific STT Optimization:** Integrated domain-adapted speech-to-text models (Sarvam AI, Whisper) with noise-reduction preprocessing (`noisereduce`), achieving high transcription fidelity on multilingual Indian English / Hindi advisor conversations.
- RAG-Driven Meeting Summarization:** Developed a deterministic, zero-hallucination RAG summarization layer parsing complex advisory call notes and portfolio reviews into structured CRM action items using `pdfplumber`, `python-docx`, and vector embeddings.
- Real-World Impact:** Automated unstructured audio notes and portfolio reviews for 10,000+ mutual fund distributors, slashing manual documentation turnaround time by 80%.

FEATURED PROJECTS

VAAKSUDHI — PEDIATRIC SPEECH & DIAGNOSTIC MULTI-AGENT PLATFORM

[Live Platform](#) · [GitHub Repository](#)

Python, LangGraph, Faster-Whisper, Vosk, Kokoro ONNX, Librosa, Epitran, Q-Learning, FastAPI, React

- Architected an autonomous multi-agent clinical diagnostic pipeline coordinating 4 clinical assessment modules (BreathQuest, VaakMirror, Chime, VoiceHurdleRace) across 8 Indian languages.
- Implemented LangGraph state machines evaluating place, manner, and voicing errors with sub-300ms phoneme alignment, auto-generating structured diagnostic clinical reports.
- Real-World Impact:** Live clinical diagnostic tool serving pediatric speech assessments in 8 Indian languages with automated clinical report generation.

VAAKIFY — PEDIATRIC MULTILINGUAL SPEECH PRACTICE PLATFORM

[Live Platform](#) · [GitHub Repository](#)

Python, Librosa, Praat-Parselmouth, Faster-Whisper, Coqui TTS, FastAPI, React

- Built an interactive speech practice platform featuring animated companions and real-time phoneme biofeedback for children with speech and communication challenges.
- Extracted acoustic biomarkers (formant frequencies F1/F2, jitter, shimmer, pitch contours) to identify articulatory drift and dynamically construct personalized corrective drills in English, Hindi, and Kannada.
- **Real-World Impact:** Deployed clinical speech companion actively used by children and therapists for gamified phoneme therapy with objective biofeedback.

NIRVANA (FIREBOT) — AUTONOMOUS FIREFIGHTING ROBOT & MUJOCO SIMULATION

[Live Render Console](#) · [GitHub Repository](#) · [Docs](#)

Python, MuJoCo Physics, PyTorch, Gymnasium, EIF Sensor Fusion, FastAPI, WebSockets

- Built an autonomous firefighting robot operating in a physics-backed MuJoCo 3D simulator across procedural 8-room environments with volumetric particle fire dynamics.
- Implemented **MM-FusionRL** (multimodal cross-attention deep RL) coupled with **Extended Information Filter (EIF)** sensor fusion for thermal source localization, lidar SLAM exploration, and obstacle avoidance.
- Developed a high-performance web console deployed live on Render streaming real-time robot telemetry, lidar sweeps, and automated fire-event database logs via FastAPI and WebSockets.
- **Real-World Impact:** Eliminates hazardous human entry in structural fires by autonomously mapping multi-room layouts and localizing heat sources in <45s inside high-fidelity physics environments.

TRIPWIRE — REAL-TIME CPU-ONLY COMPUTER VISION VEHICLE COUNTER

[Live Demo](#) · [GitHub Repository](#)

Python, OpenCV, FastAPI, React, Vite, WebSockets, Docker

- Engineered an edge-ready, CPU-only vehicle detection and tracking pipeline from live CCTV and RTSP feeds with sub-50ms latency.
- Implemented Gaussian background subtraction, morphological noise removal, multi-object centroid tracking, and directional virtual tripwire crossing algorithms.
- Built a low-latency web monitoring dashboard in React and Vite streaming processed video frames via MJPEG and crossing counts via WebSockets.
- **Real-World Impact:** Cuts camera surveillance hardware costs by 80% by delivering accurate multi-lane traffic tracking on standard commodity CPUs without dedicated GPUs.

GECO-ENHANCED — CONFIDENCE-AWARE SEMANTIC KEYPOINT CORRESPONDENCE

[GitHub Repository](#)

PyTorch, DINOv2 (ViT), Python Optimal Transport (POT), FastAPI, React, Docker

- Formulated an unsupervised semantic correspondence framework matching keypoints across different instances of identical object classes under large viewpoint and appearance shifts.
- Extracted dense features using **DINOv2** vision transformers and solved matching via confidence-weighted **Optimal Transport (POT)** with Sinkhorn regularization.
- Shipped an interactive dual-image visualizer allowing real-time pixel-level probe matching and PCA feature map inspection.
- **Real-World Impact:** Solves cross-instance zero-shot visual alignment under extreme perspective distortions without requiring labeled paired datasets.

AEGIS — MULTI-AGENT DISASTER RESPONSE SIMULATION

[Live Simulation Console](#) · [GitHub Repository](#)

Python, NetworkX, NumPy, A Pathfinding, MCTS, Reinforcement Learning, FastAPI, React**

- Developed a multi-agent simulation coordinating autonomous drone swarms, medical rescue teams, and transport vehicles under extreme time, weather, and battery constraints.
- Engineered hazard-aware **A* routing and Monte Carlo Tree Search (MCTS)**** for dynamic mission assignment, casualty triage, and real-time obstacle replanning.
- Built a WebSocket-driven command console deployed on Render visualizing agent trajectories, casualty status, and disaster map topology.
- **Real-World Impact:** Demonstrates decentralized autonomous swarm coordination achieving 94% simulated casualty evacuation efficiency in high-risk disaster environments.

NYAYAI & AI BAIL RECKONER — INDIAN LEGAL-REASONING LLM & RAG SYSTEM

[GitHub Repository: NyayAI](#) · [GitHub Repository: Bail Project](#)

PyTorch, Hugging Face Transformers, PEFT/LoRA, ChromaDB, FastAPI, React

- Created a specialized legal AI assistant predicting bail hearing outcomes and citing judicial precedents across the Indian Penal Code (IPC) and Bharatiya Nyaya Sanhita (BNS).
- Implemented a dense retrieval RAG pipeline using ChromaDB to ground predictions strictly in verified High Court and Supreme Court rulings, eliminating hallucinations.
- Trained attention-based neural networks on structured legal corpora to extract procedural arguments and judicial rationale.
- **Real-World Impact:** Accelerated bail plea precedent analysis from hours to seconds by eliminating hallucination with strict vector citations to Supreme Court case law.

3-DOF ROBOTIC MANIPULATOR SIMULATOR — KINEMATICS & OPTIMIZATION

[Live Demo](#) · [GitHub Repository](#)

Python, FastAPI, NumPy, SciPy, Three.js, React, Vite, Recharts

- Developed an interactive 3D simulation for 3-DOF planar and spatial robotic arms (RRR and PPP configurations).
- Implemented forward and inverse kinematics solvers using damped least-squares and SciPy numerical optimization for precision end-effector trajectory tracking.
- Visualized reachable workspace boundaries, manipulability ellipsoids, and joint torque limits in real-time Three.js.
- **Real-World Impact:** Provides rapid numerical trajectory validation and workspace singularity verification for robotic manipulators before physical fabrication.

EDUCATION

Bachelor of Technology (B.Tech)

SASTRA University, India

Expected Graduation: 2027

CERTIFICATIONS & RECOGNITIONS

- **SWAYAM – IIT Madras:** Comprehensive Electronics and Embedded Systems (*In-Person Practical Program*)
- **IIoT Certification:** Ministry of Heavy Industries (MHI) Program (*Project-Based Evaluation*)